

Dry Cannabis Flower for Medical use·Cannabis sativa L.·Ph. Eur. Monograph 3028·Cannabis flos

THC-dominant

Apples & Bananas

THC 15 : CBD 1

Grade IV

Lab Sample ID

Лаб. ИД на примерок

PP-QCL-26-042

Production Batch №

Производна серија №

P060052

Requested Test

Побарано испитување

Identification C — HPTLC (Param. 3)

Specification Ref.

Референца на спецификација

QCSP-IMB-001 v.02 · §3

01 Analytical Method | Аналитички метод

Method Метода	Method validation status: Cp · VERIFIED SOP · QCSOP-002 v.02
Identification Test C — HPTLC Fingerprint(Ph. Eur. 2.8.25 · mon. 3028)	
Sample preparation: 0.5 g powdered drug (355 µm) extracted with 5.0 mL methanol R by ultrasonication (15 min, 40 °C), centrifuged 5 min at 3 000 rpm, supernatant used as test solution. Reference solutions: (a) THC CRS 1.0 mg/mL in methanol R; (b) CBD CRS 1.0 mg/mL in methanol R; (c) mixed reference (a + b).	
Chromatographic conditions: HPTLC silica gel 60 F ₂₅₄ plates (Merck, 20 × 10 cm) · Application: 2 µL bands via CAMAG Linomat 5 (8 mm bandwidth, 8 mm from edge, track spacing 11 mm) · Mobile phase: toluene : diethyl ether (1:1, V/V) · Development: CAMAG ADC2 chamber, 20 min saturation, 70 mm migration distance · Derivatization: anisaldehyde-sulphuric acid reagent, heated at 100 °C / 5 min · Detection: daylight and UV 366 nm .	

02 Result(s) · HPTLC Plate Evaluation | Резултат(u)

Track	Solution Applied	Vol. (µL)	Zone & R _f Observed	Colour (Daylight)	UV 366 nm	Status
T1	Reference (a) — THC CRS	2	R _f = 0.50	Violet-blue	Red-brown fluorescence	REF
T2	Reference (b) — CBD CRS	2	R _f = 0.38	Violet-blue	Red-brown fluorescence	REF
T3	Reference (c) — Mixed THC + CBD	2	R _f = 0.50 (THC) R _f = 0.38 (CBD)	Two violet-blue zones	Two red-brown zones	SST
T4	Test solution — P060052Apples & Bananas · sample preparation as §01	2	R _f = 0.50 (THC) R _f = 0.38 (CBD)	Intense violet-blue (THC) Faint violet-blue (CBD)	Red-brown (THC) Faint red-brown (CBD)	OK
T5	Test solution — P060052 (replicate)Independent preparation from same sample	2	R _f = 0.49 (THC) R _f = 0.37 (CBD)	Intense violet-blue (THC) Faint violet-blue (CBD)	Red-brown (THC) Faint red-brown (CBD)	OK

Zone-matching assessment — THC & CBD zones conform to references CONFORMS

SST **System suitability (Track T3):** ΔR_f(THC – CBD) = 0.12 (acceptance ≥ 0.05) — **PASS**. THC and CBD zones baseline-separated. **Additional zones observed:** faint zone at R_f ≈ 0.28 (consistent with CBN/CBC minor cannabinoids) — does not interfere with THC or CBD identification. Plate development: front reached 70 mm in 18 min; symmetrical, no tailing > 1.5.

03 Method Validation · System Suitability · Measurement Uncertainty | Валидација на метод · Соодветност на системот · Мерна неодреденост

Method Status Статус на метод
Compendial · Verified for cannabis flower matrix per Ph. Eur. 5.x · Verification dossier QCV-002 v.01 · Specificity: THC and CBD zones unambiguously identified in the presence of CBN, CBC, THCA, CBDA; no false positives from excipients or matrix. Robustness: ±5 % mobile-phase ratio, ±2 min development time — no zone merging, 3-analyst concordance 100 % (κ = 1.00, n = 12 plates).
System Suitability Соодветност на системот
CAMAG Linomat 5 (S/N LIN5-PP-01 · Cal-Cert 2026-02) · Application volume accuracy ±0.1 µL · ADC2 development chamber (S/N ADC2-PP-01) · saturation pad equilibration 20 min · Plate lot: Merck 1.05641 (Lot HX98456) · Anisaldehyde spray: freshly prepared, stored < 2 h before use · Heating plate: 100 ± 2 °C (thermocouple Cal-Cert 2026-01) · Visualization: CAMAG TLC Visualizer 2 (S/N VIS2-PP-01) under daylight and UV 366 nm · Reference standards: THC CRS (Ph. Eur., Lot 4.1a) and CBD CRS (Ph. Eur., Lot 2.0).
Measurement Uncertainty Мерна неодреденост
Test C is a qualitative identity test (zone-matching) — no numerical measurement uncertainty applies. R _f values are reported for traceability; typical R _f reproducibility on silica gel 60 F ₂₅₄ plates under controlled ADC2 conditions is ±0.02 (absolute). Reliability governed by system suitability (ΔR _f ≥ 0.05), derivatization reproducibility, and reference-standard concordance.

Conforms

Result: The tested sample **conforms** to the product specification **QCSP-IMB-001 v.02 §3 (Identification C — HPTLC)** per Ph. Eur. 11.5 Monograph 3028 + general chapter 2.8.25 — principal zones in the test-solution chromatogram correspond in position (R_f), colour and fluorescence to **THC** (R_f 0.50) and **CBD** (R_f 0.38) reference zones. System suitability met ($\Delta R_f = 0.12 \geq 0.05$). QC disposition only; final batch release is certified separately by the QP in linked CoQ CoQ-PP-2026-0005.

Notes | Забелешки

- 1. These results relate **only to the tested sample(s)** as received in the QC Laboratory.
- 2. **No part of this report may be reproduced** except in full, without prior written permission of the QC Laboratory.
- 3. Measurement uncertainty is evaluated in accordance with the QC Laboratory’s quality system; for qualitative identity tests the **R_f reproducibility data and system suitability records are available on request**.
- 4. Testing was performed **in-house** by Purely Plant QC personnel under an **ISO/IEC 17025-aligned** quality system within the **MK GMP** facility, using the compendial Ph. Eur. 2.8.25 HPTLC procedure verified for the cannabis-flower matrix (QCV-002). **No analyses outsourced**. OOS handling per QCSOP-014.

Analysis Performed by

Анализата ја изврши

Analyst · QC Laboratory

Аналитичар · Лабораторија КК

Stojanka Pavlova
Lab ID PP-QCL-04

Date · Датум16.01.2026

Reviewed by

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